

# My Name Is Heyi. If I Were a Package I'd Be NumPy

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This manuscript introduces who I am, my interests, and what I hope to learn from this course.

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## 1 Who I Am

I'm Heyi Cao, a student at UW–Madison studying computer science. Although computer science is my main academic focus, many of my interests also involve visual and creative work, especially photography, aviation, and 3D design.

I am taking this course because I want to build the skills necessary for future research. In particular, I hope to learn how to approach problems in a more systematic way: identifying a question, collecting and organizing evidence, testing possible explanations, and communicating results clearly. I also want to become more comfortable with tools and workflows that support reproducible research, so that my work can be understood, checked, and built upon by others.

## 2 If I Were a Package

If I were a Python package, I would be NumPy. According to the [official NumPy documentation](#), NumPy is a fundamental package for scientific computing in Python (NumPy Developers 2026).

1. I identify with it because it provides a **foundation** for many different kinds of work:
  - data analysis
  - scientific computing
  - machine learning.
2. NumPy has also become an important foundation for scientific computing in the Python ecosystem (Harris et al. 2020). I also value building strong fundamentals and creating something that others can use as a foundation for their own work.

### **i** Why NumPy?

I like the idea of creating tools that can become a *foundation* for other people's work.

## 3 NumPy Demonstration

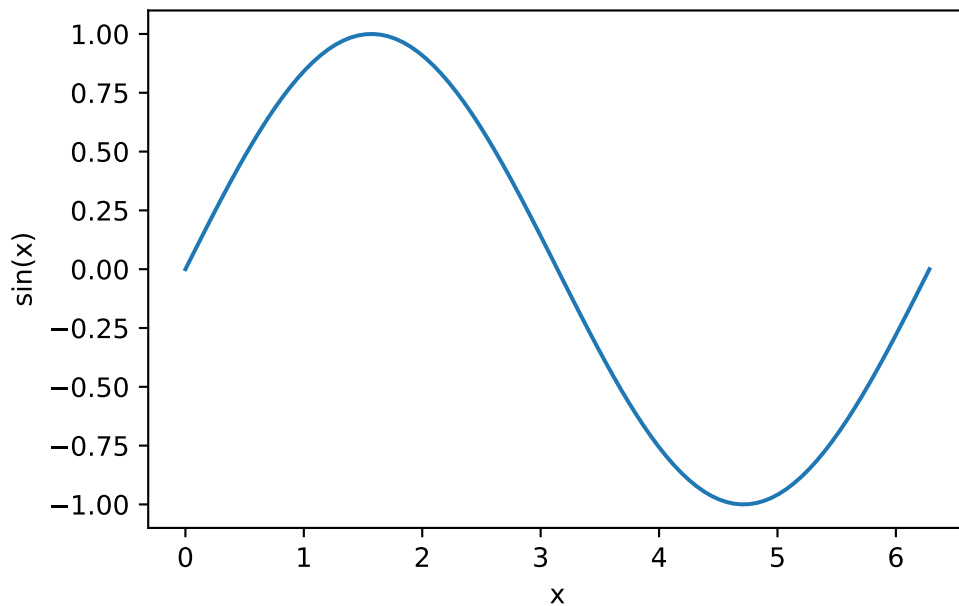


Figure 1: A sine wave generated using NumPy.

As shown in Figure 1, NumPy can generate and transform numerical data that can then be visualized clearly.

### 3.1 Working with Arrays

Table 1: Summary statistics calculated using NumPy.

|   | Statistic | Value |
|---|-----------|-------|
| 0 | Mean      | 9.4   |
| 1 | Minimum   | 4.0   |
| 2 | Maximum   | 15.0  |

Table 1 shows the max, mean, and min values of the given array.<sup>1</sup>

Harris, Charles R., K. Jarrod Millman, Stéfan J. van der Walt, et al. 2020. “Array Programming with NumPy.” *Nature* 585 (7825): 357–62. <https://doi.org/10.1038/s41586-020-2649-2>.

NumPy Developers. 2026. *What Is NumPy?* <https://numpy.org/doc/stable/user/whatisnumpy.html>.

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<sup>1</sup>The original values are not shown.